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LECTURE NOTES 1: WAVE EQUATION

WAVEGUIDES

WAVEGUIDES COMPONENTS

MICROWAVE TUBES

MICROWAVE SEMICONDUCTORS

OPTICAL FIBRE

INTRODUCTION ELECTROMAGNETIC WAVE

Wave Equation Recap

A plane wave only has dependence (and, therefore, propagation) in one direction. In the figure below, the plane wave is propagating in the z direction. Notice that the polarization is in the y direction (thus, it is $E_y(z)$), but the wave is propagating only in the z direction.

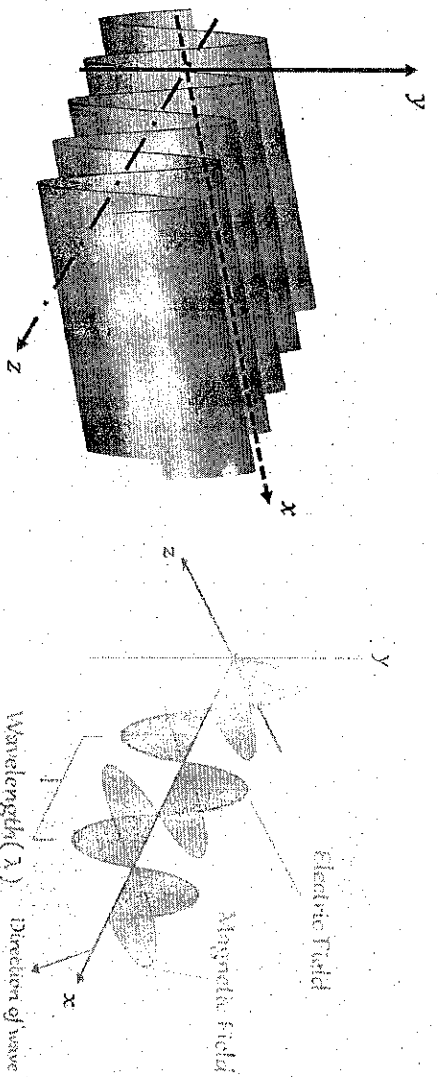


Figure 1. A plane wave $E_y(z)$ has no x or y dependence.

We define the **wavelength** λ to be the distance between successive peaks. Since the wave has no dependence in the x or y direction, we can simplify the description of the wave as follows

$$E_y(x, y, z, t) = E_y(z, t)$$

(Equ. 1)

If the wave time harmonic is known, the z and t dependence can be separated as follows:

$$E_y(z, t) = E_y(z)e^{j\omega t}$$

(Equ. 2)

Let's investigate whether (and how) this function will satisfy the wave equation. First, we'll determine the second derivative of E_y with respect to z :



$$\frac{\partial^2 E_y(z, t)}{\partial z^2} = \frac{\partial^2 E_y(z)}{\partial z^2} e^{j\omega t}$$

(Equ. 3)

Next, we'll calculate the second derivative with respect to t :

$$\frac{\partial^2 E_y(z, t)}{\partial t^2} = E_y(z) (j\omega)^2 e^{j\omega t}$$

(Equ. 4)

Substituting Equations 3 and 4 into the wave equation 2 above, we find:



$$\frac{\partial^2 E_y(z)}{\partial z^2} e^{j\omega t} - \frac{1}{c^2} E_y(z) (j\omega)^2 e^{j\omega t} = 0$$

(Equ. 5)

This can be simplified as follows:

$$\frac{\partial^2 E_y(z)}{\partial z^2} + \left(\frac{\omega}{c}\right)^2 E_y(z) = 0$$

(Equ. 6)

If we define k (called the **wave number**) as follows: as equation 7.

$$k = \frac{\omega}{c} \quad \text{(Equ. 7)}$$

Then Equation 6

simplifies to

$$\frac{\partial^2 E_y(z)}{\partial z^2} + k^2 E_y(z) = 0$$

(Equ. 8)

This is known as the **one-dimensional Helmholtz equation**, and it is a well-known differential equation with a well-known solution:

$$E_y(z) = ae^{-jkz} + be^{jkz}$$

(Equ. 9)

Adding the time dependence back into this result, we obtain:

$$E_y(z, t) = ae^{-jkz} e^{j\omega t} + be^{jkz} e^{j\omega t}$$

(Equ. 10)

Which can be written as:

$$E_y(z, t) = ae^{j(\omega t - kz)} + be^{j(\omega t + kz)}$$

(Equ. 11)

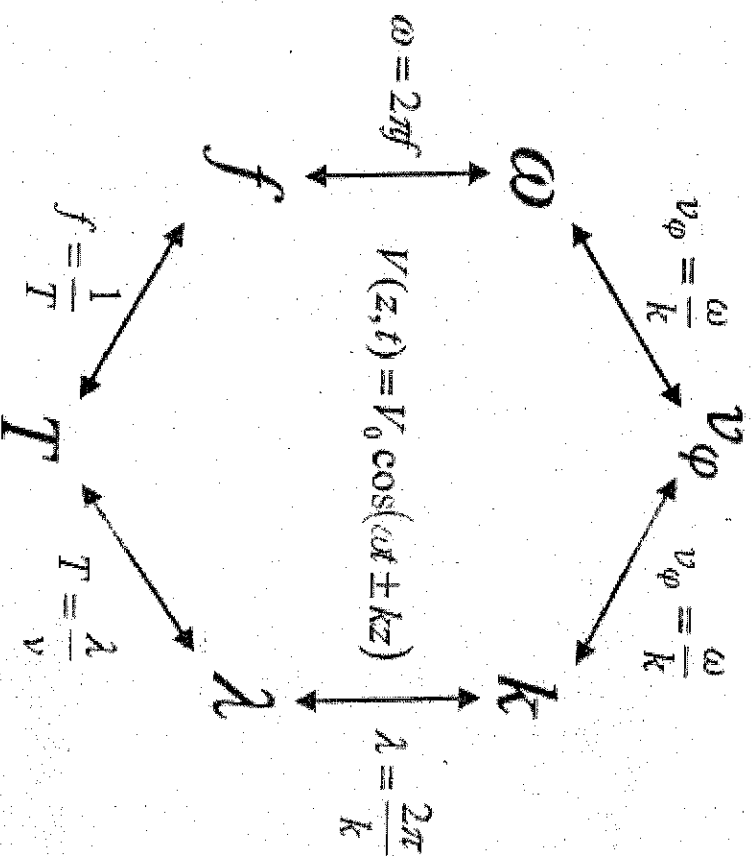


Notice that this function has the form of Equation 2 as long as Equation 1 is followed, so it is guaranteed to be a solution to the wave equation. In order to ensure that Equation 11 is a solution to the wave equation, the following relationship must be true.

$$v_{\phi} = \frac{dz}{dt} = \pm c = \pm \frac{\omega}{k}$$

(Equ. 12)

The positive signs correspond to a right-moving wave, and the negative signs to a left moving wave. We now have several variables that describe the time and spatial variation of $E(z, t)$: wavelength λ , wave number k , radial frequency ω , frequency f , period T , and phase velocity v_{ϕ} . Figure 2 below is designed to graphically demonstrate the relationships among these six variables. If you know two of the variables, you will be able to calculate the other four.



PROPAGATION THROUGH PARALLEL PLANE WAVEGUIDE

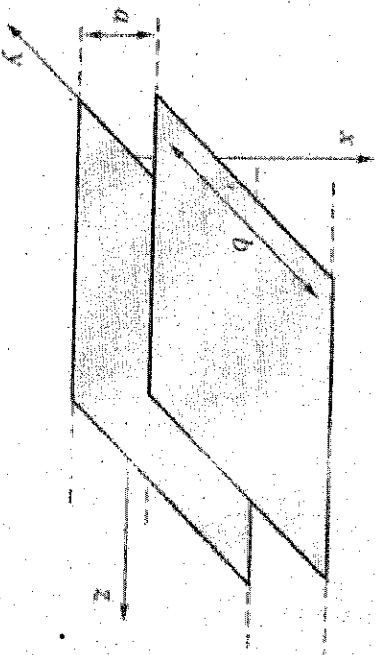


Figure 2 Parallel-plate waveguide.

In an ideal parallel-plate waveguide, the plates are assumed to be infinite in extent in the y and z directions. In practice, the width b of the guide in the y direction is finite but is typically much larger than a. The material between the plates is assumed to be a lossless dielectric.

As shown in Figure 2. The plates have an infinite extent in the z direction, in which the waves are to be guided. In practice the conductor plates would have a finite extent b in the y direction, but it is typically much larger than the plate separation a, so the effects of fringing fields at the edges are negligible. For our purposes here, we assume that the conductor plates have infinite extent ($b \gg a$) in the y direction, so the electric and magnetic fields do not vary with y. Just as in the case of waves propagating in unbounded media, the configuration and propagation of the electromagnetic fields in any guiding structure are governed by Maxwell's equations and the wave equations derived from them. However, in the case of metallic waveguides, the solutions of the wave equations are now subject to the following boundary conditions at the conductor surfaces:

$$E_{\text{tangential}} = 0$$

$$H_{\text{normal}} = 0$$

EXPRESSION FOR THE FIELD E & H VECTORS

As presented in the fig. 2 above, assuming that the medium between the perfectly

conducting plates is source-free, simple (i.e., ϵ and μ are simple constants), and lossless. (i.e., $a = 0$, μ and ϵ are real), the governing equations are the time-harmonic phasor form of Maxwell's equations, namely

$$\nabla \times \mathbf{H} = j\omega \epsilon \mathbf{E} \quad (\text{Eqn. 13.1a})$$



$$\nabla \cdot \mathbf{E} = 0$$

(Eqn. 13.11)

$$\nabla \times \mathbf{E} = -j \omega \epsilon \mu \mathbf{H}$$

(Eqn. 13.1c)

$$\nabla \cdot \mathbf{H} = 0$$

(Eqn. 13.1d)

from which the wave equations for $\mathbf{E}$ and $\mathbf{H}$ can be derived in a straightforward manner as

$$\nabla^2 \mathbf{E} - (j\beta) \mathbf{E} = 0$$

(Eqn. 14.a)

$$\nabla^2 \mathbf{H} - (j\beta) \mathbf{H} = 0$$

(Eqn. 14b)

where $\beta = \omega \sqrt{\epsilon \mu}$

FIELD SOLUTIONS FOR TE, TM, AND TEM WAVES MODES

In general, all the field components, including both E_z and H_z may need to be nonzero to satisfy the boundary conditions imposed by an arbitrary conductor or dielectric structure.

However, it is convenient and customary to divide the solutions of guided-wave equations into three categories:

$$\left. \begin{array}{l} E_z = 0 \\ H_z \neq 0 \end{array} \right\} = \text{Transverse electric (TE) waves}$$

$$\left. \begin{array}{l} E_z \neq 0 \\ H_z = 0 \end{array} \right\} = \text{Transverse magnetic (TM) waves}$$

$$E_z, H_z = 0 \longrightarrow \text{Transverse electromagnetic (TEM) waves}$$

where z is the axial direction of the guide along which the wave propagation takes place. In the following, we separately examine the field solutions for TE, TM, and TEM waves.

Transverse Electric (TE) Waves.

We first consider transverse electric (TE) waves, for which we have $E_z = 0$, $H_z \neq 0$.

With $E_z = 0$, the electric field vector of a TE wave is always and everywhere transverse to the direction of propagation. We can solve the wave equation for any of the field components; for the parallel-plate case.



Transverse Magnetic (TM) Waves.

For TM waves the magnetic field vector is always and everywhere transverse to the direction of propagation (i.e., $H_z = 0$), whereas the axial component of the electric field is nonzero (i.e., $E_z \neq 0$). We proceed with the solution in a similar manner as was done for TE waves, except that it is more convenient to use the wave equation for H_y

Transverse Electromagnetic (TEM) Waves.

Note that contrary to the TE case, the TM solutions do not all vanish for $m = 0$. Since E_z is zero for $m = 0$, the TM₀ mode is actually a transverse electromagnetic (or TEM) wave. In this case, we have TEM mode. Electric and magnetic fields between parallel planes for the TEM (TM₀) mode. Only electric field lines are shown in the top panel; the magnetic field lines are out of (or into) the page.

CUT-OFF FREQUENCY, PHASE VELOCITY, AND WAVELENGTH

Close examination of the solutions for TE and TM waves reveal a number of common characteristics: (i) E and H have sinusoidal standing-wave distributions in the x direction, (ii) any $x - y$ plane is an equiphase plane, or, in other words, the TE and TM waves are plane waves with surfaces of constant phase given by $z = c \cdot 0 \cdot n \cdot t$, and (iii) these equiphase surfaces progress (propagate) along the waveguide with a velocity $V_p = \omega / \beta$.

h is apparent that for each mode (TE or TM) identified by mode index m , there exists a critical or cut-off frequency f_{cm} (or wavelength λ_{cm} such that

$$\gamma = 0 \rightarrow f_{cm} = \frac{m V_p}{2a} = \frac{m}{2a \sqrt{\mu \epsilon}} \quad \text{or} \quad \lambda_{cm} = \frac{V_p}{f_{cm}} = \frac{2a}{m}$$

where $V_p = 1/\sqrt{\mu \epsilon}$ is the intrinsic phase velocity for uniform plane waves in the unbounded lossless medium.



Such a non-propagating mode is called an evanescent wave. For $f > f_{cm}$ the propagating wave has a wavelength $\lambda_m =$ (sometimes referred to as the guide wavelength) and phase velocity v given as

$$\begin{aligned}\lambda_m &= \frac{2\pi}{\beta_m} = \frac{\lambda}{\sqrt{1 - (f_{cm}/f)^2}} \\ v_{pm} &= \frac{\omega}{\beta_m} = \frac{v_p}{\sqrt{1 - (f_{cm}/f)^2}}\end{aligned}$$

where $\lambda = 2\pi / \beta = v / f$ is the intrinsic wavelength, being the wavelength for uniform plane waves in the unbounded lossless medium, and thus depending only on μ , ϵ , and frequency f . Note that for the TM_0 mode (which is in fact a TEM wave, as studied in

WAVEGUIDES

Generally, if the frequency of a signal or a particular band of signals is high, the bandwidth utilization is high as the signal provides more space for other signals to get accumulated. However, high frequency signals can't travel longer distances without getting attenuated. It is obvious that transmission lines help the signals to travel longer distances. Microwaves propagate through microwave circuits, components, and devices, which act as a part of Microwave transmission lines, broadly called as Waveguides.

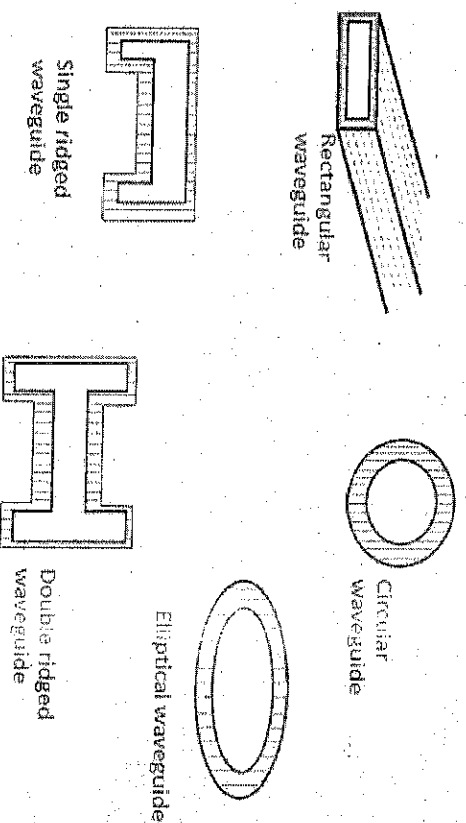
A hollow metallic tube of uniform cross-section for transmitting electromagnetic waves by successive reflections from the inner walls of the tube is called as a **Waveguide**

Types of Waveguides

There are five types of waveguides as shown and presented in the figure 6 below:

1. Rectangular waveguide
2. Circular waveguide
3. Elliptical waveguide
4. Single-ridged waveguide
5. Double-ridged waveguide

Figure 6: Types of waveguides



Waveguides, like transmission lines, are structures used to guide electromagnetic waves from point to point. However, the fundamental characteristics of waveguide and transmission line waves (modes) are quite different. The differences in these modes result from the basic differences in geometry for a transmission line and a waveguide.

Waveguides can be generally classified as either metal waveguides or dielectric waveguides. Metal waveguides normally take the form of an enclosed conducting metal pipe. The waves propagating inside the metal waveguide may be characterized by reflections from the conducting walls. The dielectric waveguide consists of dielectrics only and employs reflections from dielectric interfaces to propagate the electromagnetic wave along the waveguide.

RECTANGULAR WAVEGUIDE defined by Maxwell's Equations

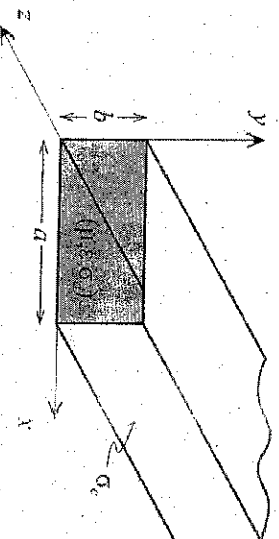


Figure 7: Rectangular waveguide

As presented earlier in the parallel plane wave guide Eqn. 13.a and 13.c earlier, given any time-harmonic source of electromagnetic radiation, the phasor electric and magnetic fields associated with the electromagnetic waves that propagate away from the source through a medium characterized by (μ, ϵ) must satisfy the source-free Maxwell's equations (in phasor form) given by

$$\begin{aligned}\nabla \times \vec{E} &= -j\omega\mu\vec{H} & (\quad) \\ \nabla \times \vec{H} &= j\omega\epsilon\vec{E} & (\quad)\end{aligned}$$

The propagation constant of a wave in a waveguide (TE or TM waves) has very different characteristics than the propagation constant for a wave on a transmission line (TEM waves). The ratio of h/k in the waveguide mode propagation constant equation can be written in terms of the cut-off frequency f_c for the given waveguide mode as follows.

$$\frac{h}{k} = \frac{h}{\omega\sqrt{\mu\epsilon}} = \frac{h}{2\pi f\sqrt{\mu\epsilon}} = \frac{f_0}{f}$$

$$f_c = \frac{h}{2\pi\sqrt{\mu\epsilon}} \quad (\text{waveguide cutoff frequency})$$

The waveguide propagation constant in terms of the waveguide cut-off frequency is

$$\gamma = jk\sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

CIRCULAR WAVEGUIDE defined by Maxwell's Equations

The same techniques used to analyze the ideal rectangular waveguide may be used to determine the modes that propagate within an ideal circular waveguide [radius = a , filled with dielectric (μ, ϵ)] The separation of variables technique yields solutions for the circular waveguide TE and TM propagating modes in terms of Bessel functions. The cut-off frequencies for the circular waveguide can be written in terms of the zeros associated with Bessel functions and derivatives of Bessel functions.

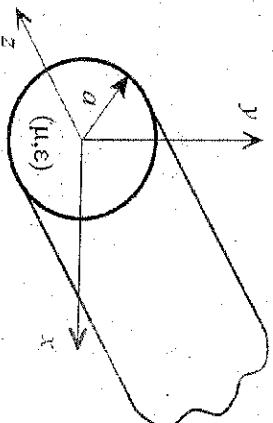


Figure 8: Circular waveguide

The cutoff frequencies of the TE and TM modes in a circular waveguide are given by

$$f_{c_{mn}}^{TE} = \frac{\chi'_{mn}}{2\pi a\sqrt{\mu\epsilon}}$$

$$f_{c_{mn}}^{TM} = \frac{\chi_{mn}}{2\pi a\sqrt{\mu\epsilon}}$$

where χ_{mn} and χ'_{mn} define the n^{th} zero of the m^{th} order Bessel function and Bessel function derivative, respectively. The values of these zeros are shown in the tables below.



TE modes

χ'_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	3.8318	1.8412	3.0542	4.2012	5.3175	6.4155	7.5013	...
$n=2$	7.0156	5.3315	6.7062	8.0153	9.2824	10.5199	11.7349	...
$n=3$	10.1735	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

TM modes

χ_{mn}	$m=0$	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	...
$n=1$	2.4049	3.8318	5.1357	6.3802	7.5884	8.7715	9.9361	...
$n=2$	5.5201	7.1056	8.4173	9.7610	11.0647	12.3386	13.5893	...
$n=3$	8.6537	10.1735	11.6199	13.0152	14.3726	15.7002	17.0038	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Note that the dominant mode in a circular waveguide is the TE_{11} mode, followed in order by the TM_{01} mode, the TE_{21} mode and the TE_{01} mode.

Example (Circular waveguide)

Design an air-filled circular waveguide yielding a frequency separation of 1 GHz between the cut-off frequencies of the dominant mode and the next highest mode.

The cut-off frequencies of the TE_{11} mode (dominant mode) and the TM_{01} mode (next highest mode) for an air-filled circular waveguide are

$$f_{c11}^{TE} = \frac{\chi'_{11}}{2\pi a \sqrt{\mu_o \epsilon_o}} = \frac{1.8412}{2\pi a \sqrt{\mu_o \epsilon_o}}$$

$$f_{c01}^{TM} = \frac{\chi_{01}}{2\pi a \sqrt{\mu_o \epsilon_o}} = \frac{2.4049}{2\pi a \sqrt{\mu_o \epsilon_o}}$$

For a difference of 1 GHz between these frequencies, we write

$$f_{c01}^{TM} - f_{c11}^{TE} = \frac{2.4049 - 1.8412}{2\pi a \sqrt{\mu_o \epsilon_o}} = 10^9$$

Solving this equation for the waveguide radius gives:



$$\alpha = \frac{2.4049 - 1.8412}{2\pi 10^9 \sqrt{\mu_o \epsilon_o}} = 2.6896 \text{ cm}$$

The corresponding cut-off frequencies for this waveguide are

$$f_{c11}^{TE} = \frac{\chi'_{11}}{2\pi a \sqrt{\mu_o \epsilon_o}} = \frac{1.8412}{2\pi (0.026896) \sqrt{\mu_o \epsilon_o}} = 3.266 \text{ GHz}$$

$$f_{c01}^{TM} = \frac{\chi_{01}}{2\pi a \sqrt{\mu_o \epsilon_o}} = \frac{2.4049}{2\pi (0.026896) \sqrt{\mu_o \epsilon_o}} = 4.266 \text{ GHz}$$

COMPARISON OF WAVEGUIDE AND TRANSMISSION LINE CHARACTERISTICS

TRANSMISSION LINE	WAVEGUIDE
1. Two or more conductors separated by some insulating medium (two-wire, coaxial, microstrip, etc.).	Metal waveguides are typically one enclosed conductor filled with an insulating medium (rectangular, circular) while a dielectric waveguide consists of multiple dielectrics.
2. Normal operating mode is the TEM or quasi-TEM mode (can support TE and TM modes but these modes are typically undesirable)	Operating modes are TE or TM modes (cannot support a TEM mode).
3. No cut-off frequency for the TEM mode. Transmission lines can transmit signals from DC up to high frequency.	Must operate the waveguide at a frequency above the respective TE or TM mode cut-off frequency for that mode to propagate.
4. Significant signal attenuation at high frequencies due to conductor and dielectric losses.	Lower signal attenuation at high frequencies than transmission lines.
5. Small cross-section transmission lines (like coaxial cables) can only transmit low power levels due to the relatively high fields concentrated at specific locations within the device (field levels are limited by dielectric breakdown).	Metal waveguides can transmit high power levels. The fields of the propagating wave are spread more uniformly over a larger cross-sectional area than the small cross-section transmission line.
6. Large cross-section transmission lines (like power transmission lines) can transmit high power levels.	Large cross-section (low frequency) waveguides are impractical due to large size and high cost.

PRINCIPLE OF WAVEGUIDE COMPONENTS

Match Loads

Match loads is simply impedance matching and in general, this can be achieved by using a waveguide iris as shown in figure below.

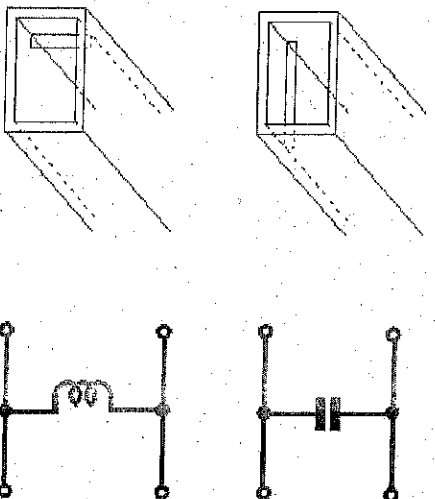


Figure 9: Impedance matching using a waveguide iris

The waveguide iris may either be on only one side of the waveguide, or there may be a waveguide iris on both sides to balance the system. A single waveguide iris is often referred to as an asymmetric waveguide iris or diaphragm and one where there are two, one either side is known as a symmetrical waveguide iris.

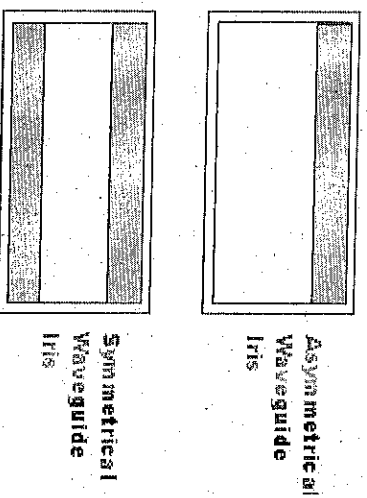


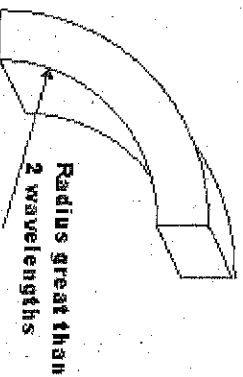
Figure 10: Symmetrical and asymmetrical waveguide iris implementations

A combination of both E and H plane waveguide irises can be used to provide both inductive and capacitive reactance. This forms a tuned circuit. At resonance, the iris acts as a high impedance shunt. Above or below resonance, the iris acts as a capacitive or inductive reactance.

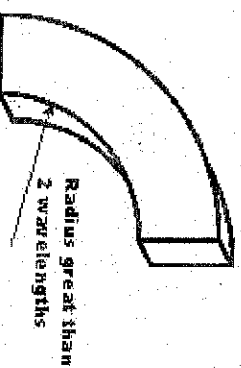
Waveguide Bends

Usually, a waveguide bend allows changes in the direction of the transmission and these bends are categorized into either waveguide E bend or H bend depending on the field it distorted. There are several ways in which waveguide bends can be accomplished which, may be used according to the applications and the requirements as listed and shown below.

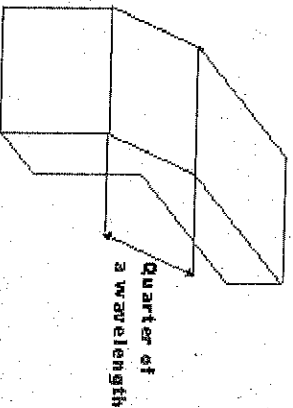
Waveguide E bend



Waveguide H bend



Waveguide sharp E bend



Waveguide sharp H bend

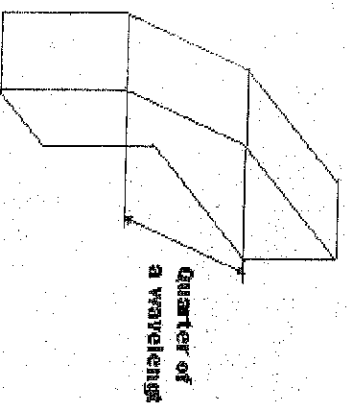


Figure 11: Waveguide BENDS

Each type of bend is achieved in a way that enables the signal to propagate correctly and with the minimum of disruption to the respective fields and hence to the overall signal. Both waveguide E or H bend are form of waveguide bend are very similar to another, except that the individual bend distorts the corresponding E or H field.

To prevent reflections in these waveguides, however, the corresponding (i.e., E and H) bends must have a radius greater than two wavelengths.

Attenuators

In order to control power levels in a microwave system by partially absorbing the transmitted microwave signal, attenuators are employed. Resistive films (dielectric glass slab coated with aquadag) are used in designing of both fixed and variable attenuators.

A co-axial fixed attenuator uses the dielectric lossy material inside the center conductor of the co-axial line to absorb some of the center conductor microwave power propagating through it dielectric rod decides the amount of attenuation introduced. The microwave power absorbed by the lossy material is dissipated as heat

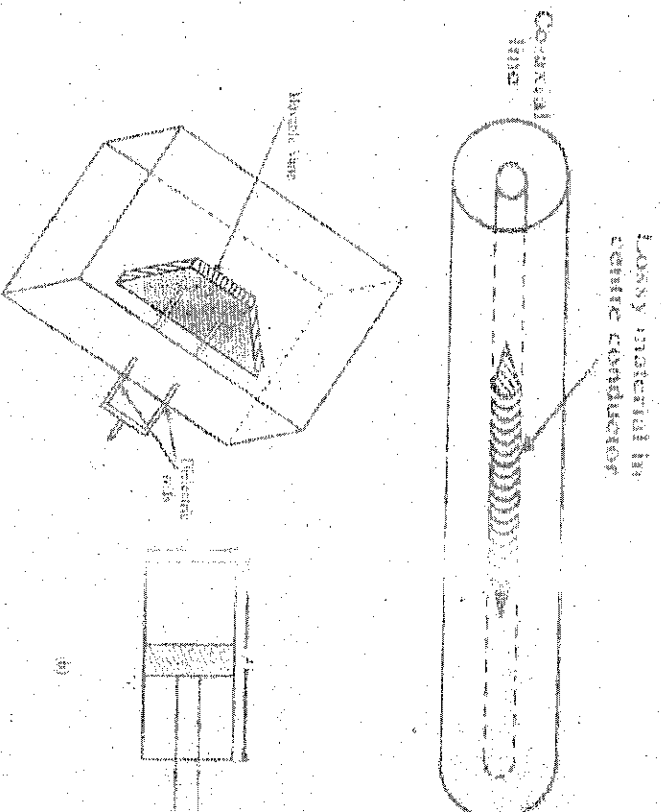
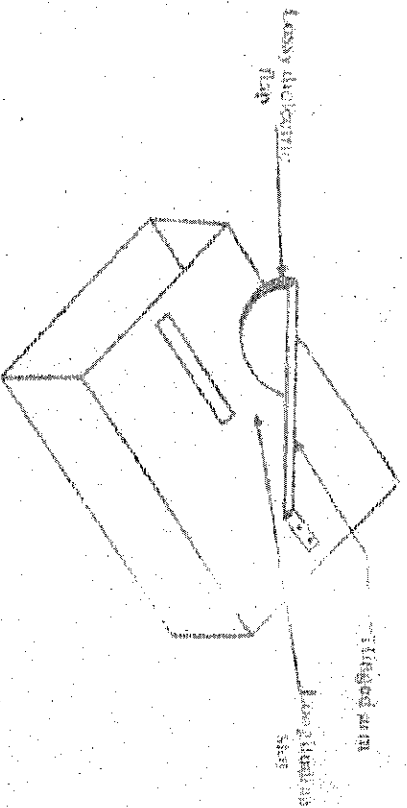


Figure 12: (a) Waveguide Attenuators

In waveguides, the dielectric slab coated with aduadag is placed at the center of the waveguide parallel to the maximum E-field for dominant TE_{10} mode. Induced current on the lossy material due to incoming microwave signal, results in power dissipation, leading to attenuation of the signal. The dielectric slab is tapered at both ends up to a length of more than half wavelength to reduce reflections as shown in figure 12. The dielectric slab may be made movable along the breadth of the waveguide by supporting it with two dielectric rods separated by an odd multiple of quarter guide wavelength and perpendicular to electric field.

Figure 12: (b) Waveguide Attenuators



Phase Shifters:

A microwave phase shifter is a two-port device which produces a variable shift in phase of the incoming microwave signal. A lossless dielectric slab when placed inside the rectangular waveguide produces a phase shift.

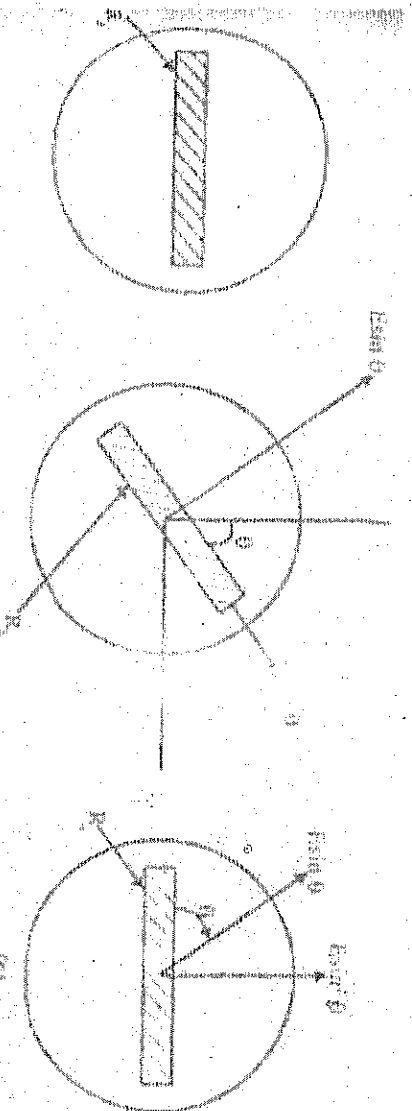


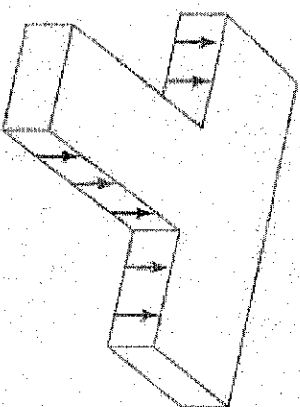
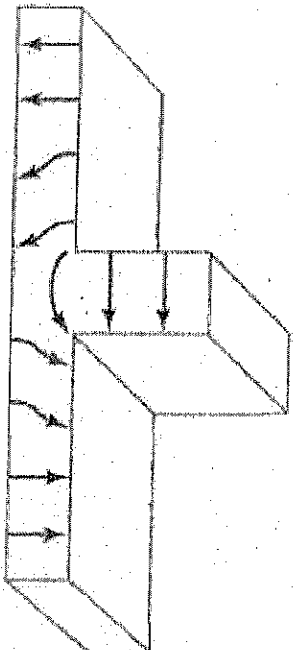
Figure 13: (b) Waveguide Phase Shifters

Multiport Junctions

T-Junction Power Divider: The T-junction power divider is a 3-port network that can be constructed either from a transmission line or from the waveguide depending upon the frequency of operation. For very high frequency, power divider using waveguide includes the following types: E-PLANE TEE, E-PLANE TEE, and E-H PLANE TEE or MAGIC TEE as presented in the figures below

E-PLANE TEE

H-PLANE TEE:



E-H PLANE TEE/MAGIC TEE

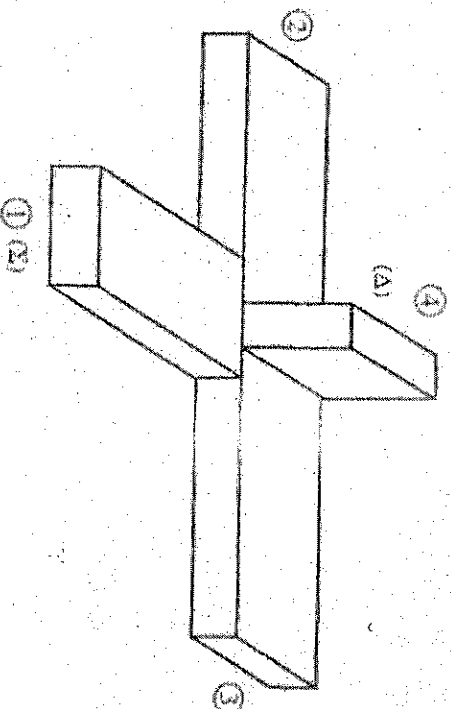


Figure 14: Waveguide Multiport junctions

Reading Assignment I

Read, understand, sketch, and label the following waveguide components:

- i. Couplers
- ii. Isolators
- iii. Duplexers and resonant cavities.

MICROWAVE TUBES

Microwave tubes are a type of vacuum tube that generate and amplify high frequency microwave signals from 300 MHz to 300 GHz. They can generate high output power levels from a few hundred watts to more than 10 MW. Microwave tubes generally are classified into two major categories namely the linear beam tubes (O-Types) and the cross-field tubes (M-Types tubes). Microwave tubes are commonly used in Radio Detecting and Ranging (RadAr) systems such as military radar, civilian radar-like weather detection, highway collision avoidance, electronic warfare, etc. As the most commonly used microwave tubes, classification of linear beam tubes is shown in the figure below.

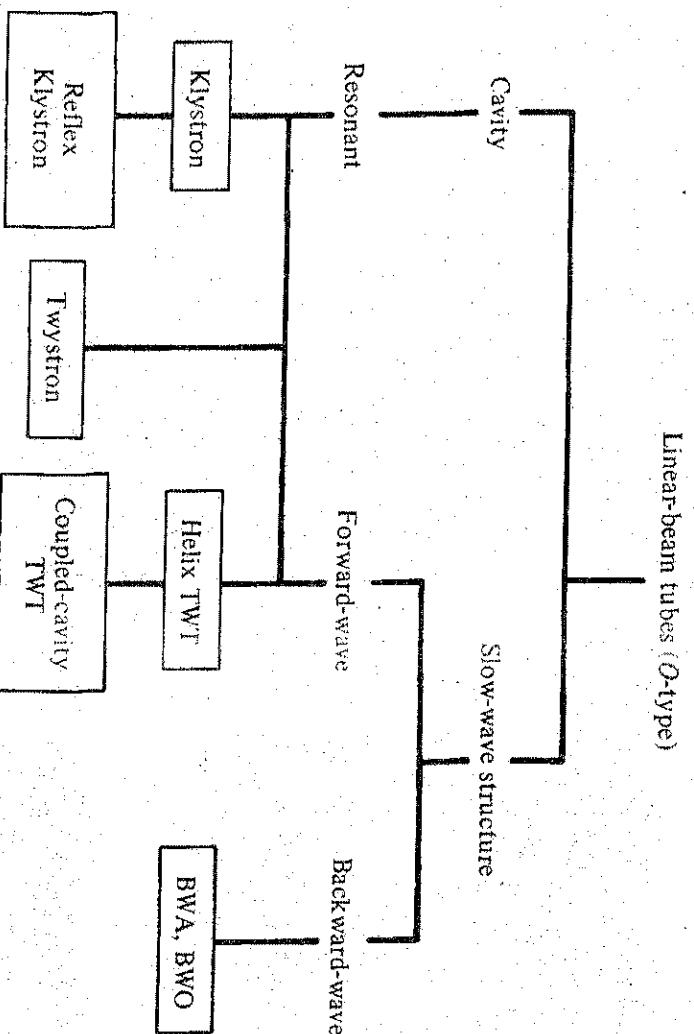


Figure 16: Classification of linear beam tubes.

Energy Transfer Mechanism

In most of the microwave tubes, the signal is placed in a cavity gap and electrons are forced to cross the gap at time when they face maximum opposition. Crossing the gap under opposition lead to transfer of energy to the cavity gap signal. When the gap voltage is sinusoidal time-varying and the charge crossing is continuous and uniform, which is usually the case, no net transfer of energy takes place between cavity and the charge crossing the gap. It is because the energy transfer is equal and opposite in direction during a half cycle



when compared to half cycle resulting in no net transfer of energy in a cycle.

To have net energy transfer, preferably maximum, from electron beam to gap signal voltage the distributed charge is compressed into a thin sheet or bunch, so that it requires less time to cross the gap and it is arranged such that the bunch crossing is at peak gap voltage so that the bunch faces maximum opposition and retardation from the signal voltage.

When the gap voltage is sinusoidal and bunch crossing is at a uniform and constant rate, for maximum unidirectional flow of energy, there is only one instant, either at positive peak or negative peak, for the bunch to cross the gap. The bunch crossing hence must be once per cycle of the gap voltage. In case of bunch crossing at a uniform rate of f , transfer of maximum energy can take place only with a component of grid gap fields whose frequency is also f . Other components of the grid gap voltage like $2f$, $4f$, $8f$, etc., do not involve in the energy transfer, whereas the components $3f$, $5f$, $6f$, etc., and $f/2$, $f/3$, $f/4$, etc., the transferred amount of energy is negligible.

Klystron

Klystron is another type of linear-beam tube. In this case the electrons flow in a straight line and a magnetic field is used to focalize the electron beam. They work on the principle of velocity modulation. Velocity modulation is the variation in the velocity of an electron beam when electrons in the beam are being accelerated and slowed down. This variation is normally achieved by varying the velocity of the beam, which causes the electrons to bunch up and generate some amount of RF current. Klystrons typically applies the transit-time effect principle by varying the velocity of an electron beam to form an electron bunch. The resonant cavities in the tube modulate the electric field along the axis. Depending on the number of the resonant cavities, klystrons are categorized into two-cavity klystrons, multicavity klystrons, and reflex klystrons.

Two Cavity Klystron

The two-cavity klystron is a widely used microwave amplifier operated by the principles of velocity and current modulation. It consists of an electron gun, focussing and accelerating grids, two identical cavities separated by a distance and at the far end a grounded collector

plate. The electron gun emits electrons from the surface of its cathode and then they are focussed into a beam. Using a dc accelerating positive voltage the beam is accelerated to high velocities.

Characteristics of two cavity klystron

Efficiency:	about 40%
Power output in CW mode:	up to 500 kW at 10 GHz
Power output in Pulsed mode:	up to 30 MW at 10 GHz
Power gain:	about 30 dB

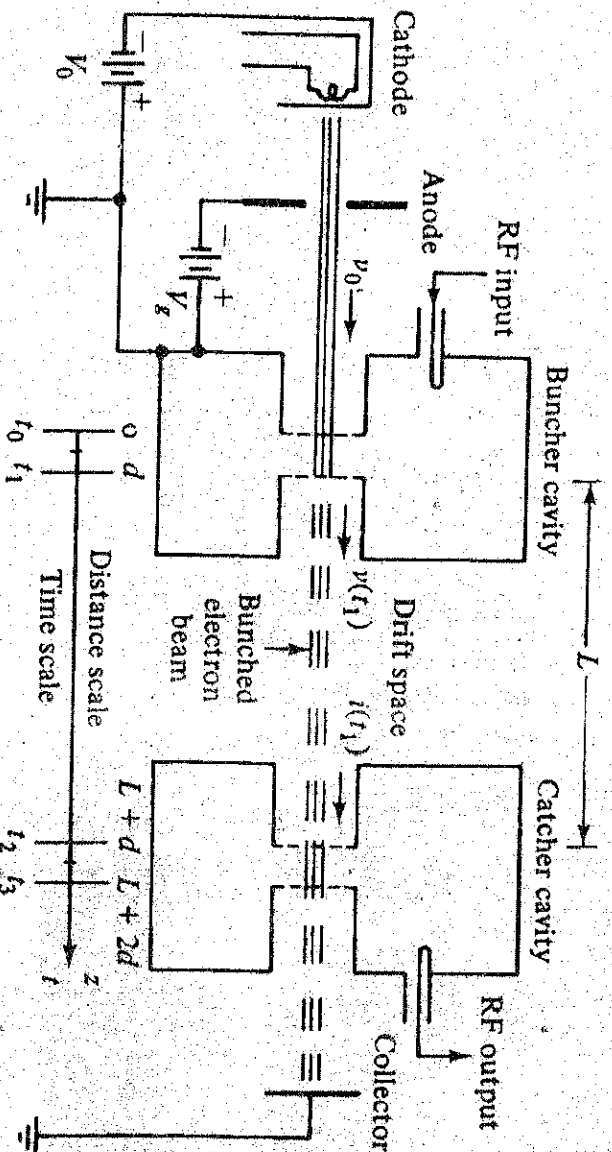


Figure 17: (a) The two-cavity klystron

Components of two cavity klystron

1. **Cathode:** Source of electrons
2. **Anode:** for formation of electron beam
3. **Buncher Cavity:** A reentrant type resonator cavity which is kept at a +ve voltage of V_0 with respect to cathode to effect acceleration of electrons. RF input voltage of $V_0 \sin \omega t$ is applied to buncher cavity.
4. **Catcher cavity:** Similar to buncher cavity. The amplified output signal $V_0 \sin \omega t$ is obtained from this cavity.

5. **Collector:** The electrons after transfer of energy to catcher cavity are collected by the collector.

Definition of Parameters

The parameters used in the description and operation of two cavity klystron includes:

$V_0 = D C$ voltage between cathode and buncher cavity.

$V_1 =$ Amplitude of input RF signal, $V_1 < V_0$

$\omega = 2\pi f =$ Input signal angular frequency. It is also equal to resonant frequency of both the cavities.

$v_2 =$ Uniform velocity of electrons between cathode and buncher cavity.

$t_2 =$ Time at which electrons enter the buncher cavity $t_1 =$ Time at which electrons leave the buncher cavity

$T =$ Transit time of electrons in the buncher cavity $= t_1 - t_2$

$\theta_s =$ Angle /Phase variation of input signal during the transit time $= \omega T$

$\beta =$ Beam Coupling Coefficient of the buncher / Catcher Cavity.

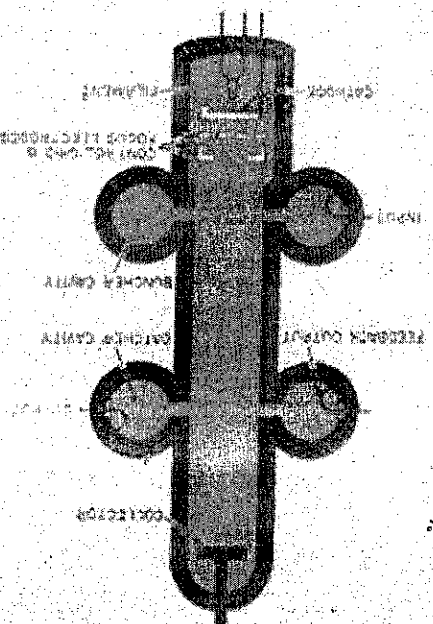


Figure 17: (b) The klystron

When the electrons enter the buncher cavity with uniform velocity V_0 interact with the field due to input RF signal $V_1 \sin \omega t$. The time varying field in the cavity cause the electrons to accelerate or decelerate and there by electrons undergo velocity modulation.

Let $(t) =$ Velocity of electrons at $t = t_1$ at the output of buncher cavity

- This is a time varying quantity
- Refer fig-5.3 for velocity modulation of electrons



Let d = (cavity width of buncher) / (catcher cavity)

L = spacing between buncher and catcher cavities

NOTE THAT: L is the design parameter for optimum performance of the klystron amplifier

t_1 = Time at which electrons enter the catcher cavity

t_4 = Time at which electrons leave the catcher cavity

$$r = \frac{d}{v_0} = t_1 - t_0$$

Because $V_1 < V_0$

Evaluation of $v(t_1)$

(t_1) is the instantaneous velocity of electrons which is a time varying quantity and primarily depends upon the average voltage in the gap during the time „ t “ i.e., during the time period (transit) the electrons are influenced by the field.

V_{avg} = Average voltage in the gap during time t



$$V_{avg} = \frac{1}{\tau} \int_{t_0}^{t_1} V_i \sin \omega t \, dt$$

$$= \frac{V_i}{\omega \tau} [\cos \omega t]_{t_0}^{t_1}$$

$$V_{avg} = \frac{V_i}{\omega \tau} [\cos \omega t_1 - \cos \omega t_0] \quad V_{avg} = \frac{V_i}{\omega \tau} [\cos \omega t_0 - \cos \omega t_1]$$

$$\text{Where } \tau = t_1 - t_0$$

$$t_1 = t_0 + \tau = t_0 + \frac{d}{v_0}$$

$$V_{avg} = \frac{V_i}{\omega \tau} \left[\cos \omega t_0 - \cos \left(\omega t_0 + \frac{\omega d}{v_0} \right) \right]$$

$$\text{Where } \theta_g = \omega \tau = \frac{\omega d}{v_0} \quad (5.3)$$

$$V_{avg} = \frac{V_i}{\omega \tau} [\cos \omega t_0 - \cos(\omega t_0 + \theta_g)]$$

$$\text{Let } A = \omega t_0 + \frac{\theta_g}{2}$$

$$\text{and } B = \frac{\theta_g}{2}$$

$$\text{Since } \cos(A-B) - \cos(A+B) = 2 \sin A \sin B$$

$$V_{avg} = V_i \left\{ \frac{\sin(\omega d / 2 v_0)}{\omega d / 2 v_0} \right\} \sin \left(\omega t_0 + \frac{\omega d}{2 v_0} \right)$$

$$= V_i \frac{\sin \theta_g / 2}{\theta_g / 2} \sin \left(\omega t_0 + \frac{\theta_g}{2} \right)$$

β_1 = beam coupling coefficient of input (buncher) cavity by definite

$$\beta_1 = \frac{\sin \theta_g / 2}{\theta_g / 2} \quad (5.4)$$

$$V_{avg} = V_i \beta_1 \sin \left(\omega t_0 + \frac{\theta_g}{2} \right)$$

$$\text{As we have seen earlier } v_0 = \sqrt{\frac{2eV}{m}}$$

$$|||y u(t_0) = \sqrt{\frac{2e}{m}} \sqrt{V_i \beta_1 \sin \left(\omega t_0 + \frac{\theta_g}{2} \right) + V_0}$$

$$= \sqrt{\frac{2e}{m} V_0 \left[1 + \frac{\beta_1 V_i}{V_0} \sin \left(\omega t_0 + \frac{\theta_g}{2} \right) \right]}$$



Since $V_1 \ll V_0$, $\frac{\beta V_1}{V_0} \ll 1$

Using binomial expansion $\sqrt{1+x} = 1 + \frac{x}{2}$ for $x \ll 1$

$$w(t_1) = v_0 \left[1 + \frac{\beta V_1}{2V_0} \sin \left(\omega t_0 + \frac{\theta}{2} \right) \right]$$

Since $\tau = t_1 - t_0$

$$\theta_g = \omega \tau = \omega t_1 - \omega t_0$$

$$\omega t_0 = \omega t_1 + \omega t_0$$

$$\omega t_0 + \theta_g / 2 = \omega t_1 - \theta_g + \theta_g / 2 = \omega t_1 - \theta_g / 2$$

$$w(t_1) = v_0 \left[1 + \frac{\beta V_1}{2V_0} \sin \left(\omega t_1 - \frac{\theta_g}{2} \right) \right] \quad (5.5)$$

$$v(t_1) = v_0 \left[1 + \frac{\beta V_1}{2V_0} \sin \left(\omega t_0 + \frac{\theta_g}{2} \right) \right] \quad (5.6)$$

Numerical : The parameters of a 2 cavity klystron are

$$V_0 = 1000 \text{ V}, \quad I_0 = 25 \text{ mA (beam current)}$$

$$d = 1 \text{ mm}, \quad f = 3 \text{ GHz}$$

Find out β_1 beam coupling coefficient

$$\begin{aligned} \text{Solution } v_0 &= 0.593 \times 10^6 \sqrt{V_0} = 0.593 \times 10^6 \times \sqrt{1000} \\ &= 1.88 \times 10^7 \text{ m/s} \end{aligned}$$

$$\theta_g = \frac{\omega d}{v_0} = \frac{2\pi \times 3 \times 10^9 \times 10^{-3}}{1.88 \times 10^7} = 1.002 \text{ rad}$$

$$\beta_1 = \frac{\sin \theta_g / 2}{\theta_g / 2} = \frac{\sin 0.5}{0.5} = 0.958$$

Reading Assignment II

- I. Describe and further evaluate the related parameters associated with microwave tubes: Magnetron.
- II. States any two of its applications

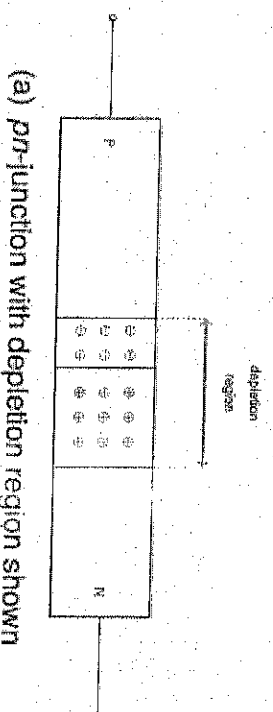


Magnetron

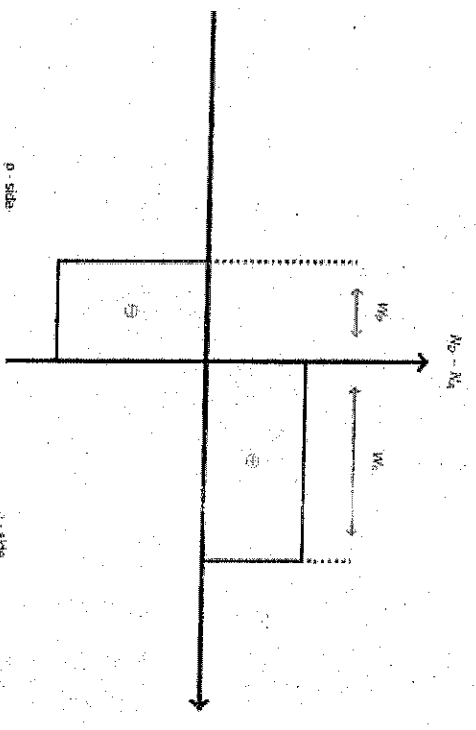
The magnetron is cross field microwave tube. These tubes are normally circular in appearance. In magnetrons, the electric field and magnetic field run perpendicular to each other. The crossed electric and magnetic fields are used to generate high output power, which in turn is used in applications such as radar equipment. Magnetrons combine a vacuum tube with cavity resonators and a powerful magnet. A magnetron consists of an electron gun/cathode that emits the electrons and multiple anode cavities. A permanent magnet is placed on the backside of cathode and the space between the anode cavity and the cathode is called interacting space. There are two main types of magnetrons; pulsed magnetron and CW magnetron. Pulsed magnetrons are used in radars and CW ones are used in applications like microwave ovens.

MICROWAVE SEMI-CONDUCTOR DIODES

PN-junction Diodes: Semiconductor materials: Silicon-Germanium.



(a) *pn*-junction with depletion region shown



(b) *pn*-junction charge density profile

Figure 18: Simplified representation of a *pn*-junction



- When two semiconductor materials are brought into contact, the Fermi levels must become aligned so as to be the same throughout the crystal.
- Surplus electrons from the n -region will diffuse into the p -region leaving a region of net positive charge in the n -region, near the junction. Similarly, surplus holes from the p -region will diffuse into the n -region leaving a region of net negative charge in the p -region, near the junction.
- The region immediately either side of the junction will now have been depleted of majority carriers, and is therefore referred to as the *depletion region*, also referred to as the *space charge region*, as illustrated in figure 3.
- With no external applied voltage, a potential will exist across the pn -junction due to the build-up of carriers on each side. This is known as the *built-in potential*.
- When a forward bias is applied externally, the depletion region shrinks as negative charge carriers are repelled from the negative terminal towards the junction and holes are repelled from the positive terminal towards the junction. This reduces the energy required for charge carriers to cross the depletion region. As the applied voltage increases, current starts to flow across the junction once the applied voltage reaches the *Barrier potential*

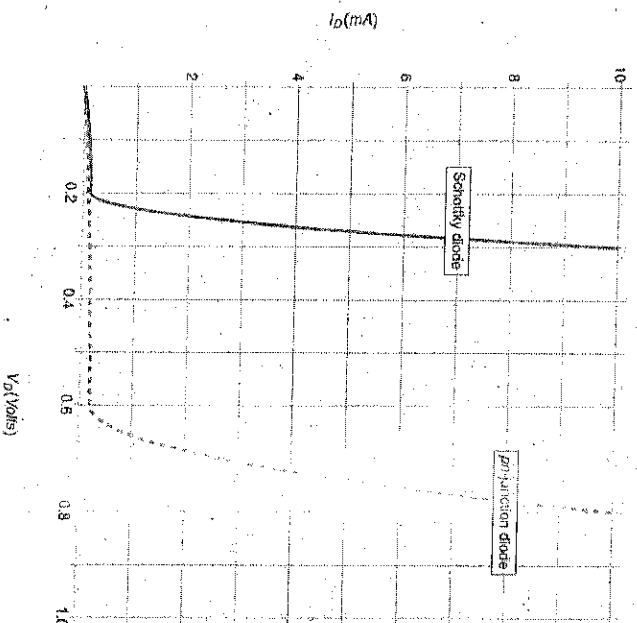


Figure 19: Schottky diode vs pn -junction forward bias characteristics (Silicon)



Schottky Diode

- Schottky Diodes employ a Metal-Semiconductor Contact Junction. There is no *p-n* junction, as such.
- Schottky Diodes have very fast switching times due to their small capacitance and the fact that they are *majority carrier* devices.
- Schottky diodes have a very short reverse *recovery time*. For *p-n* junctions, the reverse recovery time is between 5 to 100 ns. For a Schottky diode it is normally between 5 and 100 ns.
- Schottky diodes are widely used in RF circuits as mixers and detectors.

Gunn diode

- Simply a slab of N-type GaAs (Gallium Arsenide) with no PN junctions
- The band structure of certain compound semiconductors, such as GaAs and InP, has two local minima in the conduction band: one where the electrons have a low effective mass and a high mobility and a second local minimum at a higher energy level where electrons have a higher effective mass and a lower mobility.
- This causes a concentration of free electrons called a domain which moves through the device from Cathode to Anode.

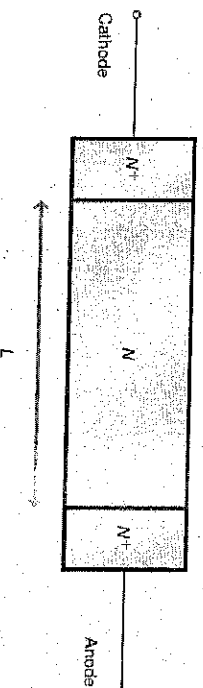


Figure 20: Gunn diode construction

Gunn Diode Characteristic

The drop in electron mobility with increasing electric field means that a sample of this material will exhibit a decrease in current with increasing applied voltage, that is to say a negative differential resistance.

At higher voltages, the normal increase of current with voltage relation resumes once the bulk of the carriers are kicked into the higher energy-mass valley. Therefore, the negative resistance only occurs over a limited range of voltages, as illustrated.

OPTICAL FIBER

An optical fiber is a cylindrical dielectric waveguide made of low-loss materials such as silica glass. It has a central core in which the light is guided, embedded in an outer cladding of slightly lower refractive index (Fig. 8.0-1). Light rays' incident on the core-cladding boundary at angles greater than the critical angle undergo total internal reflection and are guided through the core without refraction. Rays of greater inclination to the fiber axis lose part of their power into the cladding at each reflection and are not guided.

Fiber optic principles are essentially the same as those that apply in planar dielectric waveguides, except for the cylindrical geometry. In both types of waveguide light propagates in the form of modes. Each mode travels along the axis of the waveguide with a distinct propagation constant and group velocity, maintaining its transverse spatial distribution and its polarization.

One of the difficulties associated with light propagation in multimode fibers arises from the differences among the group velocities of the modes. This results in a variety of travel times so that light pulses are broadened as they travel through the fiber. This effect, called modal dispersion, limits the speed at which adjacent pulses can be sent without overlapping and therefore the speed at which a fiber-optic communication system can operate.

Modal dispersion can be reduced by grading the refractive index of the fiber core from a maximum value at its center to a minimum value at the core-cladding boundary. The fiber is then called a graded-index fiber, whereas conventional fibers

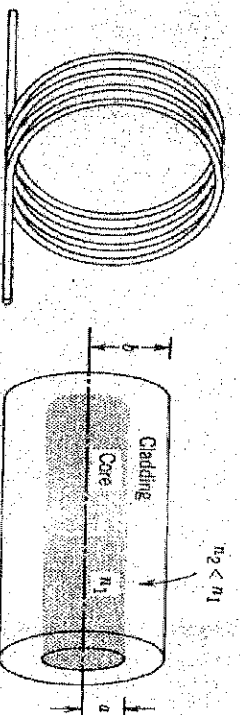


Figure 21: An optical fiber.

Structure and Physics of an Optical Fiber

Generally, an optical fibers used in communications have a very simple structure. They consist of two sections: the glass core and the cladding layer as shown in figure xx below. The core is a cylindrical structure, and the cladding is a cylinder without a core. Core and cladding have different refractive indices, with the core having a refractive index, n_1 , which is slightly higher than that of the cladding, n_2 . It is this difference in refractive indices that enables the fiber to guide the light. Because of this guiding property, the fiber is also referred to as an "optical waveguide." As a minimum there is also a further layer known as the secondary cladding that does not participate in the propagation but gives the fiber a minimum level of protection. This second layer is referred to as a coating.

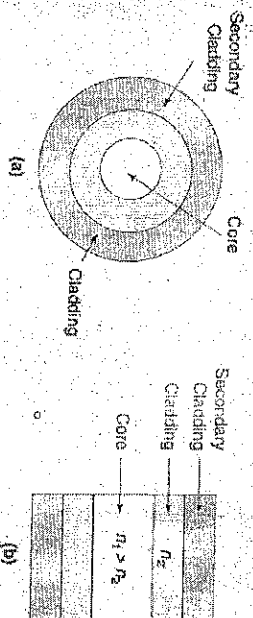
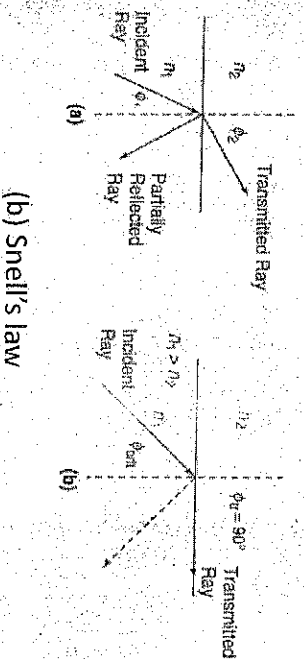


Figure 22: (a) Cross section and (b) longitudinal cross section of a typical optical fiber.



(b) Snell's law

The basics of light propagation can be discussed with the use of geometric optics. The basic law of light guidance is Snell's law. Consider two dielectric media with different refractive indices and with $n_1 > n_2$ and that are in perfect contact, as shown in Snell's law Figure above. At the interface between the two dielectrics, the incident and refracted rays satisfy Snell's law of refraction — i.e.,



$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$
$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1$$

In addition to the refracted ray there is a small amount of reflected light in the medium with refractive index n_1 . Because $n_1 > n_2$ then always $\theta_1 > \theta_2$. As the angle of the incident ray increases there is an angle at which the refracted ray emerges parallel to the interface between the two dielectrics Figure 22 (b).

This angle is referred to as the critical angle, θ_{crit} , and from Snell's law is given by

$$\sin \theta_{crit} = \frac{n_2}{n_1}$$

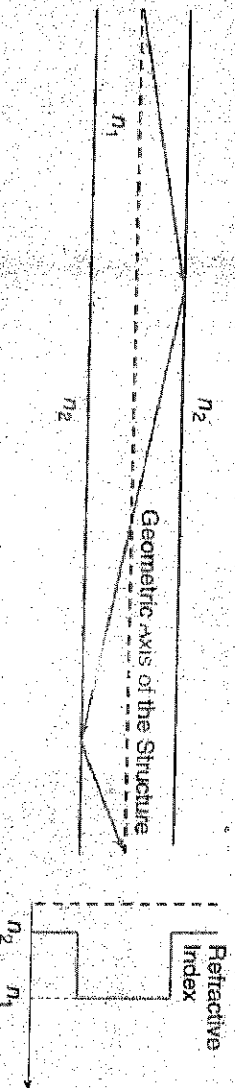


Figure 23: Light guidance using Snell's law.

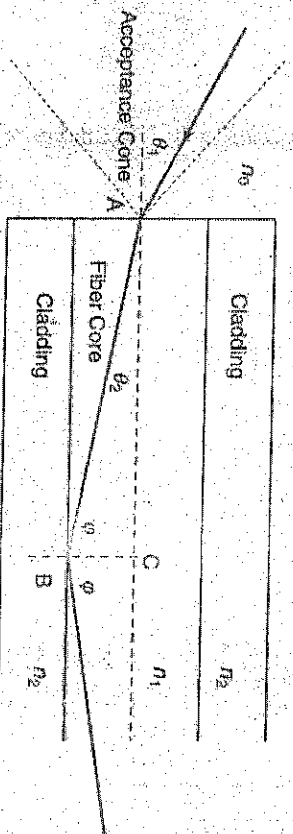


Figure 24: Geometry for the derivation of the acceptance angle.

If the angle of the incident ray is greater than the critical angle, the ray is reflected back into the medium with refractive index n_1 . This basic idea can be used to propagate a light ray in a structure with $n_1 > n_2$ and Figure 23 illustrates this idea. The light ray incident at an angle

greater than the critical angle can propagate down the waveguide through a series of total reflections at the interface between the two dielectrics.

Figure 24 illustrates the geometry for the derivation of the acceptance angle. To satisfy the condition for total internal reflection, the ray arriving at the interface, between the fiber and outside medium, say air, must have an angle of incidence less than θ_{acc} otherwise the internal angle will not satisfy the condition for total reflection, and the energy of the ray will be lost in the cladding.

Consider that a ray with an incident angle less than the θ_{acc} , say θ_1 , enters the fiber at the interface of the core (n_1) and the outside medium, say air (n_0), and the ray lies in the meridional plane. From Snell's law at the interface, we obtain

$$n_0 \sin \theta_1 = n_1 \sin \theta_2$$

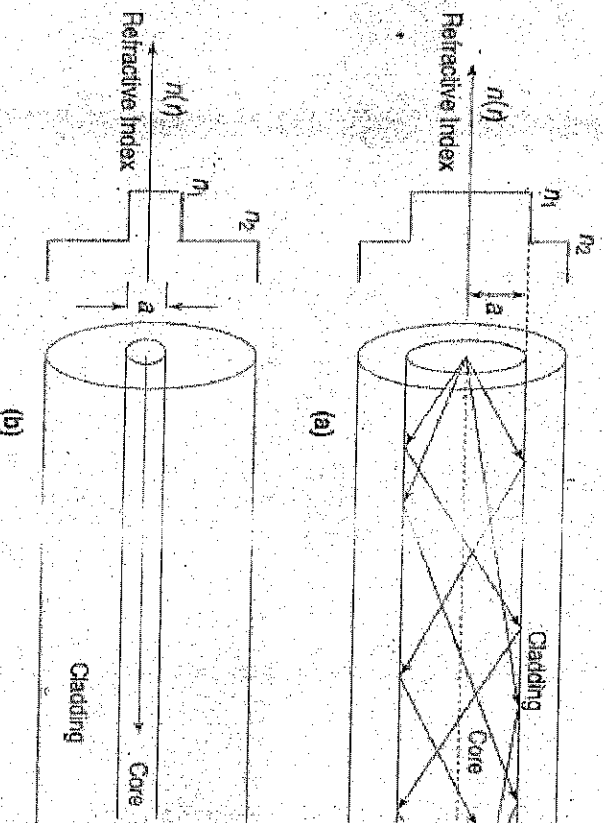


Fig. 25 (a) Multimode and (b) single-mode propagation for step index fiber.



LECTURE NOTES 2:

RADAR

RADAR EQUATION

RADAR FREQUENCIES

RADAR APPLICATIONS

DOPPLER RADAR SYSTEM

RADAR PROBLEMS

RADAR

The word *Radar* is an Acronym for **Radio Detection And Ranging**

Radar is a detection system that uses radio waves to determine the range, angle, or velocity of objects. It can be used to detect aircraft, ships, spacecraft, guided missiles, motor vehicles, weather formations, and terrain.

A radar system consists of a transmitter producing electromagnetic waves in the radio or microwaves domain, a transmitting antenna, a receiving antenna (often the same antenna is used for transmitting and receiving) and a receiver and processor to determine properties of the object(s). there are two basic radar systems

1. Monostatic radar systems
2. Bistatic radar systems

Radar block diagram

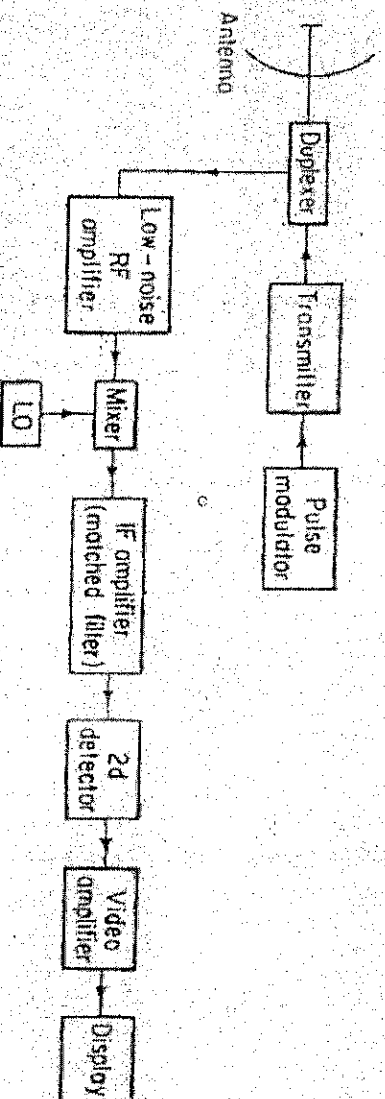


Figure 26: The block diagram of a monostatic Radar.

The system is called monostatic Radar because the system uses a single transceiver i.e., same antenna is used for transmission and reception. There are two sections of radar: The transmitter section and the receiver section Transmitter section

- **Transmitter:** the transmitter may be a power amplifier such as klystron, travelling wave tube or transistor amplifier. This will generate the Electrical energy at R.F. (Radio Frequency).
- **Pulse modulator:** The power amplifier (Such as klystron, TWT) produces a highpower signal, may be in terms of megawatts. Pulse modulator shown in the block is used as a switch, which will turn on and off the power amplifier.



- **Threshold decision:** The output of video amplifier is given to the threshold detector where it is decided whether the received signal is from a target or just because of the presence of noise.
- **Display:** The Display is generally a CRT (Cathode Ray Tube) (a) 'A' scope (b) PPI
 - i) 'A' scope provided Range and Echo power.
 - ii) PPI measures Range and bearing (azimuth angles) iii) In addition there are other displays like 'B' scope, 'D' scope etc.

General Working Principle

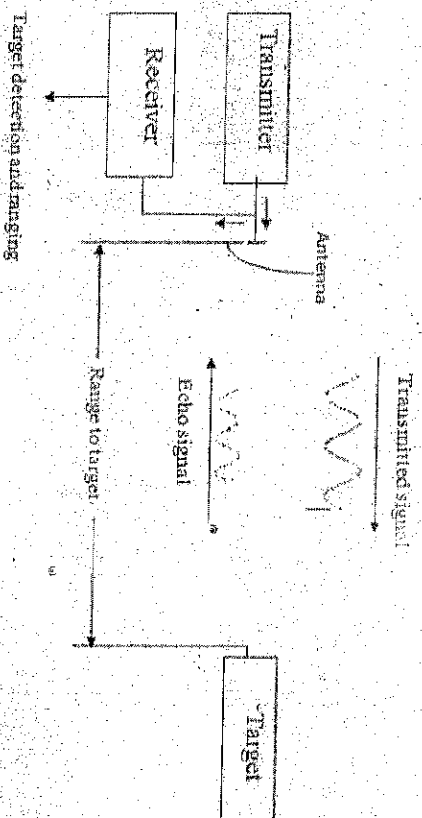


Fig. 27 Working principle of a RADAR system.

Functions Of Radar

- Detects the presence of target
- Gives the range of the target from the Radar station
- Gives the azimuth angle and elevation angle of the target
- Gives the radial velocity of target.

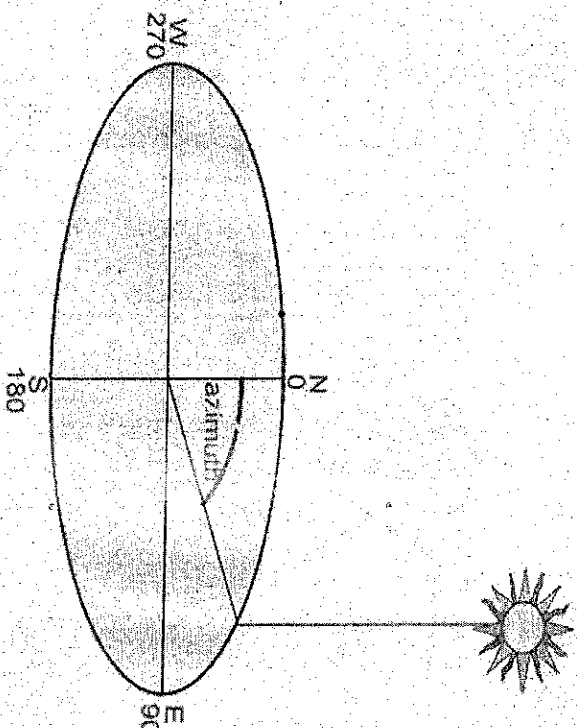


Fig. 28 RADAR coordinates and Ranges

RADAR System Definition of TERMS

When measuring a RADAR system Range, some terms/acronyms defined below are commonly used:

P R F : The no of radar pulses transmitted per second is known as pulse repetition frequency or pulse repetition rate (FF).

P R T : The time from beginning of first pulse to the beginning of the next is called pulse repetition time (T),

$$F_p = \frac{1}{T_p}$$

T_R Time taken by EM pulse to travel to target and come back to same antenna

R Range of target

C Velocity of EM waves $= 3 \times 10^8$ Meters/sec

Rest Time or Receiver Time:

The time between two successive transmitted pulse is called as Rest Time or Receiver Time

Radar Range Determination

- The most common radar waveform is a train of narrow, rectangular-shape pulses modulating a sine wave carrier.
- The distance, or range, to the target is determined by measuring the time T_R taken by the pulse to travel to the target and return.
- Electromagnetic energy in free space travels with the speed of light c (i.e. 3×10^8 m/s) therefore range R is given by :



$$R = c \times T_R / 2$$

- The range R in kilometers or nautical miles, and T_R in microseconds, the above relation becomes: $(k m) = 0.15 \times T_R (\mu S)$ or
 $R (n m i) = 0.081 \times T_R (\mu S)$
- Each microsecond of round-trip travel time corresponds to 0.081 nautical mile, **0.093** statute mile, **150** meters, **164** yards, or **492** feet.
- (1 mile = 0.8689 nautical mile or 1.6 km 1nauticalmile= 1.15078 miles or 1.8412km)
- It takes 12.35 μs for radar signal to travel a nautical mile and back

RADAR FREQUENCIES

- RF spectrum is very scarce and as such Radars are allotted only a certain frequency bands for their operation by International Telecom Union ITU
- During 2nd world war, to keep the secrecy, certain code words were used. The same designations are continued even today
- Lima Band (L) 1GHZ-2GHZ, Sierra band(S) 2GHZ-4GHZ, Charlie Band (C) 4GHZ-8GHZ, Xera Band (X) 8GHZ-12GHZ
- ITU (International Telecommunication Union) allocated a portion of these bands for Radar

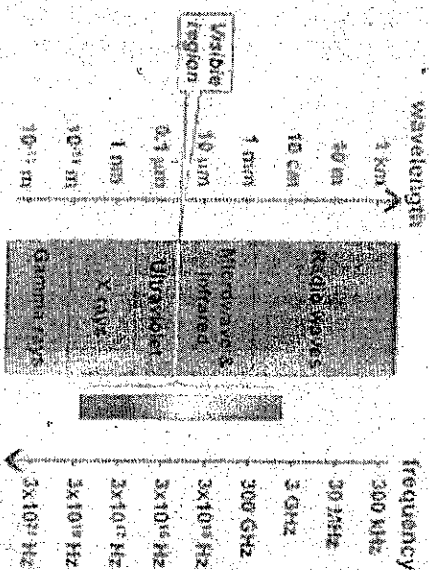


Fig. 29 Electromagnetic spectrum Wavelength and corresponding frequencies.

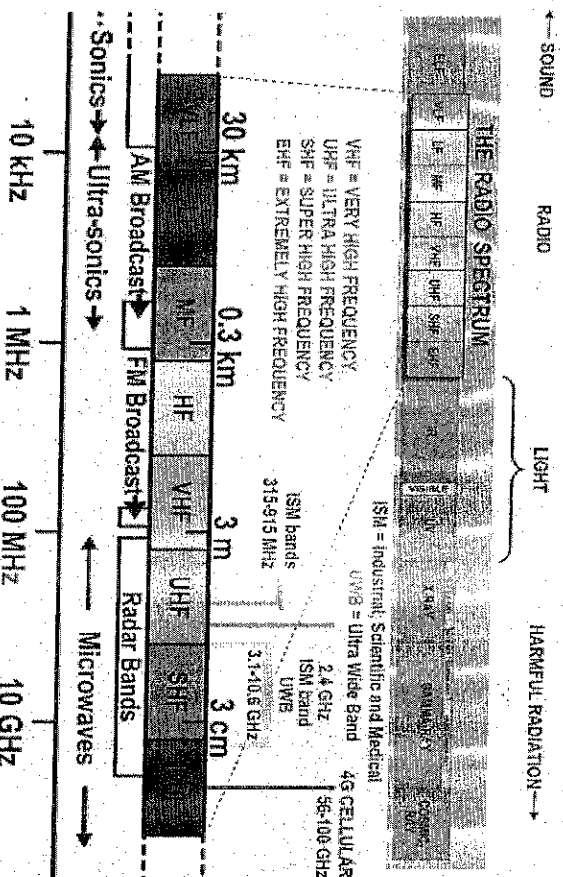


Fig. 29 (a) Electromagnetic Radio spectrum

Fig. 29 (c) Electromagnetic spectrum allocations for RADAR based on ITU

FIELDS OF APPLICATION OF RADAR SYSTEM

The following includes the application of a RADAR

- Military
- Remote Sensing



- c. Air Traffic Control
- d. Law Enforcement and Highway
- e. Security
- f. Aircraft Safety and Navigation
- g. Ship Safety
- h. Space
- i. Miscellaneous Applications

Military:

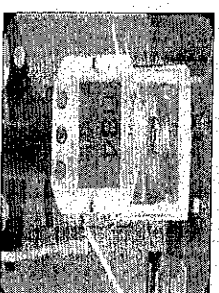
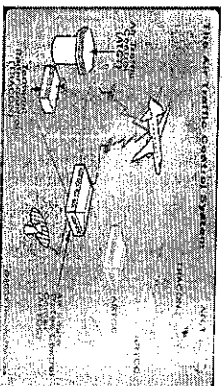
- Important part of air defense system, operation of offensive missiles & other weapons.
- Target detection, target tracking & weapon control.
- Also used in area, ground & air surveillance.

Air Traffic Control

- Used to safely control air traffic in the vicinity of the airports and enroute.
- Ground vehicular traffic & aircraft taxing.
- Mapping of regions of rain in the vicinity of airports & weather.

Law Enforcement & Highway Safety:

- Radar speed meters are used by police for enforcing speed limits.
- It is used for warning of pending collision, actuating air bag or warning of obstruction or people behind a vehicle or in the side blind zone



Remote Sensing

- Weather observation-
- T.V. Reporting
- Planetary observation
- Below ground probing



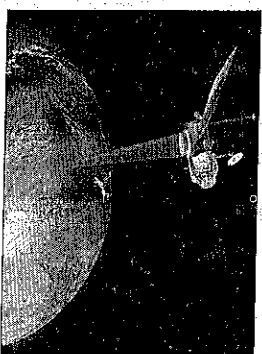
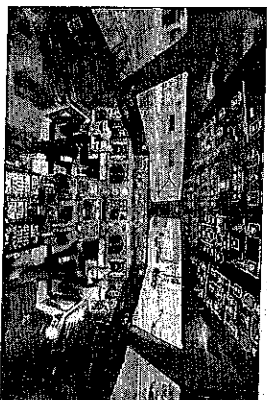
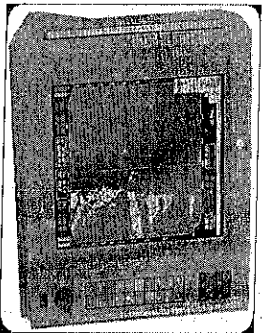
- Mapping of sea ice

Aircraft Safety & Navigation

- Low flying military aircrafts rely on terrain avoidance & terrain following radars to avoid collision with high terrain & obstructions

Ship Safety

- Radar is found on ships & boats for collision avoidance & to observe navigation buoys, when the visibility is poor.
- Shore based radars are used for surveillance of harbours & river traffic.



Space

- Space vehicles have used radar for clocking & for landing on the moon.
- Used for planetary exploration.
- Ground based radars are used for detection & tracking of satellites & other space objects.
- Used for radio astronomy.

Other APPLICATIONS

- It is used for in industry for the non-contact measurement of speed & distance.
- Used for oil & gas exploration.
- Used to study movements of insects & birds.



Prediction Of Range Performance

The simple form of the radar equation derived earlier expresses the maximum radar range R_{max} in terms of radar and target parameters:

$$R_{max} = [(P_t \times G \times A_e \times \sigma) / (4\pi)^2 \times S_{min}]^{1/4}$$

where

P_t = transmitted power, measured in watts

G = antenna gain

A_e = antenna effective aperture, measured in m^2

σ = radar cross section, measured in m^2

S_{min} = minimum detectable signal, measured in watts

All the parameters are to some extent under the control of the radar designer, except for the target cross section σ .

Doppler Effect

- Doppler effect implies that the frequency of a wave when transmitted by the source is not necessarily the same as the frequency of the transmitted wave when picked by the receiver.
- The received frequency depends upon the relative motion between the transmitter and receiver.
- If transmitter and receiver both are moving towards each other the received frequency higher, this is true even one is moving.
- If they are moving apart the received signal frequency decreases and if both are stationary, the frequency remains the same. This change in frequency is known as Doppler shift.
- Doppler shift depends upon the relative velocity between radar and target



RADAR EQUATION AND PROBLEMS

Reading Assignment IV